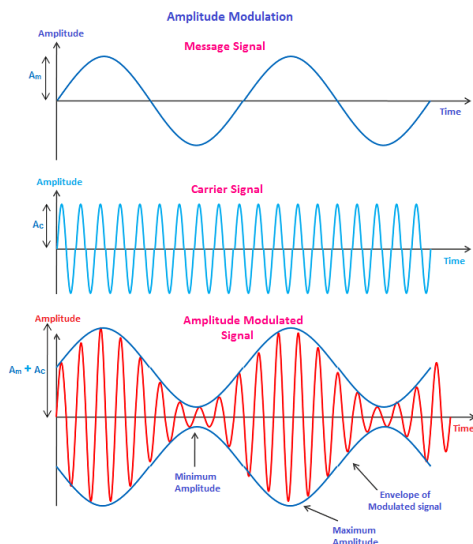


ELEX 2220 – Lab 14 Electronics Project

Margareth Sabate – A01380070 – Set A

Project Description:

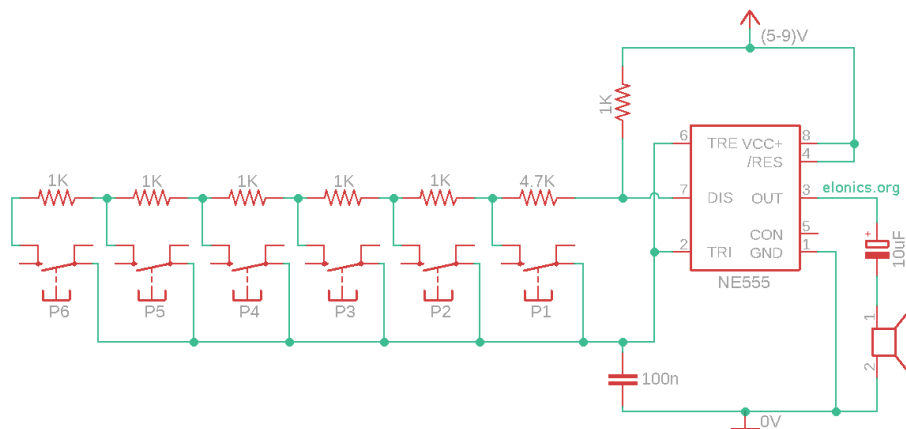
My project is a piano circuit that uses 555 timers in astable mode to generate different sound frequencies. I use pushbuttons to change resistor values at R2, which adjusts the output frequency and creates different musical notes. This works because of how frequency relates to pitch, with higher frequencies producing higher-pitched sounds. I also added an SPDT switch to shift the octave up or down by switching between two capacitor values at C1, either halving or doubling the frequency.



I tried to include a sound modulation feature to make the notes sound more robotic. I use amplitude modulation, where I combine a 100 Hz modulating signal with the piano signal using an AND gate.

This turns the output on and off quickly, changing the texture of the sound. I added a switch to choose between normal and modulated sound, but the modulation didn't fully work and ended up glitchy.

Piano Circuit from: <https://elonics.org/electric-piano-circuit-using-555-ic/>



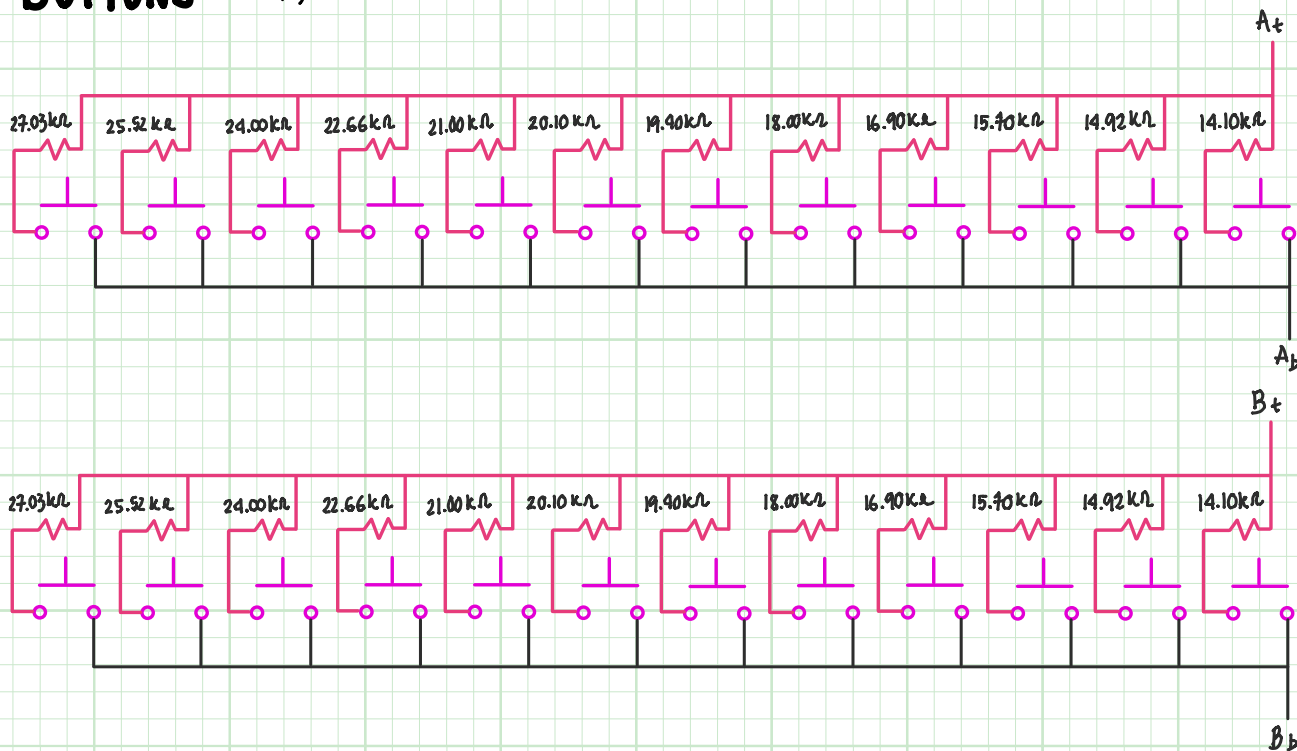
ELECTRIC PIANO CIRCUIT

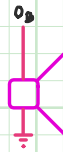
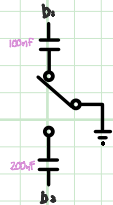
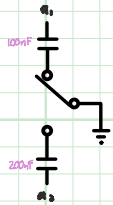
Bill of Materials (BoM)

Circuit w/o Modulation			Circuit w/ Modulation		
Part Number	Description	Quantity	Part Number	Description	Quantity
NE555	Timer IC	3	NE555	Timer IC	3
SW-TAC-6MM	6mm Tactile Pushbutton Switch, Momentary	24	74HC08	Quad 2 input AND Gate	1
100nF	Capacitor	2	SW-TAC-6MM	6mm Tactile Pushbutton Switch, Momentary	24
200nF	Capacitor	2	100nF	Capacitor	3
10kΩ	Resistor	18	200nF	Capacitor	2
22kΩ	Resistor	8	10μF	Capacitor	2
4.7kΩ	Resistor	20	10kΩ	Resistor	18
3.3kΩ	Resistor	4	22kΩ	Resistor	8
2.2k	Resistor	2	4.7kΩ	Resistor	20
1kΩ	Resistor	10	3.3kΩ	Resistor	4
330Ω	Resistor	6	2.2k	Resistor	2
220Ω	Resistor	4	1kΩ	Resistor	11
100Ω	Resistor	2	330Ω	Resistor	6
SPK-2IN-8OHM-0.5W	2" 8Ω 0.5W Speaker	2	220Ω	Resistor	5
SW-SPDT-GRAYHILL	Grayhill SPDT Switch, 1 Pole Double Throw	2	100Ω	Resistor	2
			SPK-2IN-8OHM-0.5W	2" 8Ω 0.5W Speaker	2
			SW-SPDT-GRAYHILL	Grayhill SPDT Switch, 1 Pole Double Throw	4

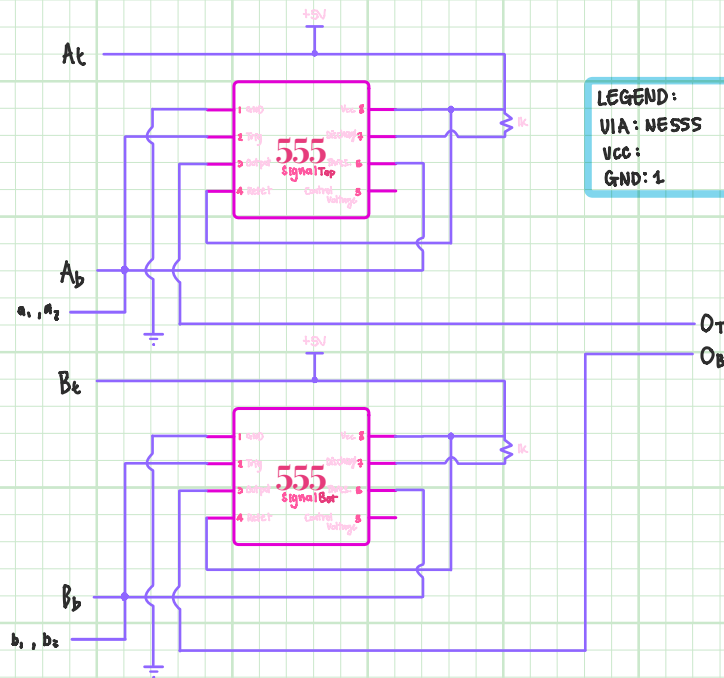
Note	Target Freq (Hz)	Target R2 (kΩ)	R2 Combo Used	Actual Freq (Hz)	Error (%)
C4	261.63 Hz	26.48 kΩ	330 Ω + 4.7 kΩ + 22 kΩ = 27.03 kΩ	261.53 Hz	-0.04
C#4	277.18 Hz	24.50 kΩ	220 Ω + 3.3 kΩ + 22 kΩ = 25.52 kΩ	276.71 Hz	-0.17
D4	293.66 Hz	22.85 kΩ	1 kΩ + 1 kΩ + 22 kΩ = 24.00 kΩ	293.88 Hz	+0.07
D#4	311.13 Hz	21.51 kΩ	330 Ω + 330 Ω + 22 kΩ = 22.66 kΩ	310.88 Hz	-0.08
E4	329.63 Hz	20.04 kΩ	1 kΩ + 10 kΩ + 10 kΩ = 21.00 kΩ	334.88 Hz	+1.59
F4	349.23 Hz	19.35 kΩ	100 Ω + 10 kΩ + 10 kΩ = 20.10 kΩ	349.51 Hz	+0.08
F#4	369.99 Hz	18.22 kΩ	4.7 kΩ + 4.7 kΩ + 10 kΩ = 19.40 kΩ	361.81 Hz	-2.21
G4	392.00 Hz	17.12 kΩ	3.3 kΩ + 4.7 kΩ + 10 kΩ = 18.00 kΩ	389.19 Hz	-0.72
G#4	415.30 Hz	16.02 kΩ	2.2 kΩ + 4.7 kΩ + 10 kΩ = 16.90 kΩ	413.79 Hz	-0.36
A4	440.00 Hz	14.86 kΩ	1 kΩ + 4.7 kΩ + 10 kΩ = 15.70 kΩ	444.44 Hz	+1.01
A#4	466.16 Hz	13.78 kΩ	220 Ω + 4.7 kΩ + 10 kΩ = 14.92 kΩ	466.93 Hz	+0.16
B4	493.88 Hz	12.77 kΩ	4.7 kΩ + 4.7 kΩ + 4.7 kΩ = 14.10 kΩ	493.15 Hz	-0.15

BUTTONS (Piano keys)

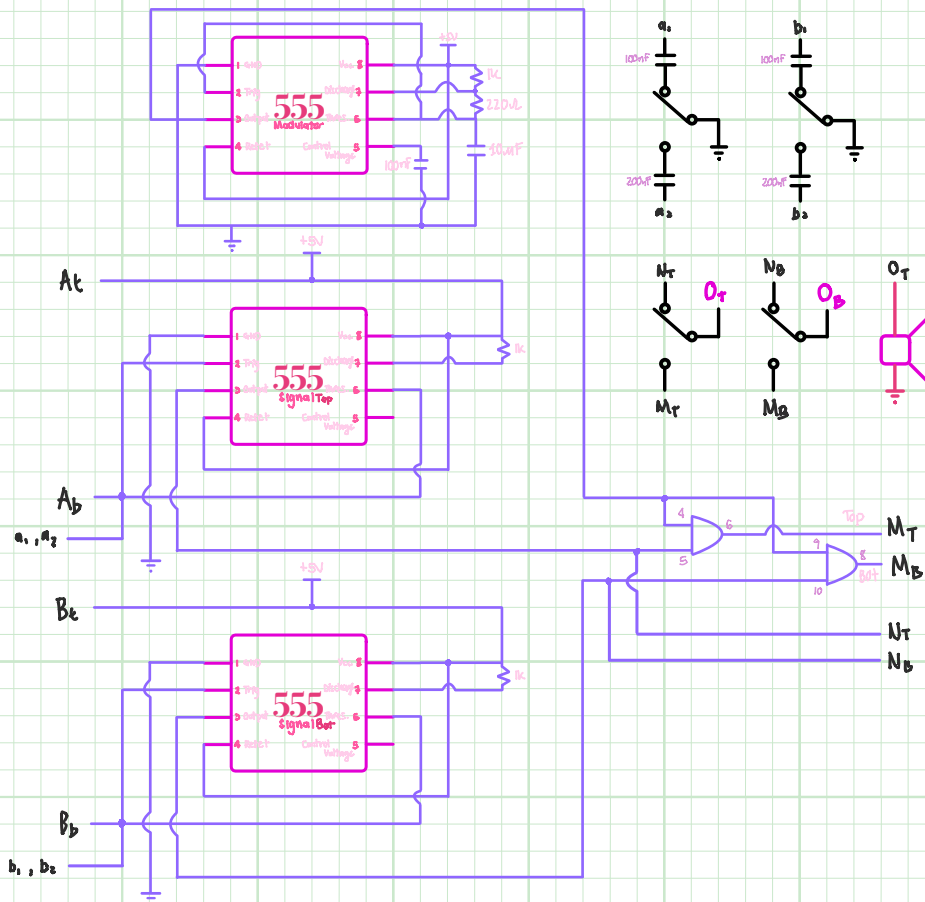




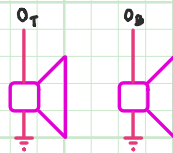
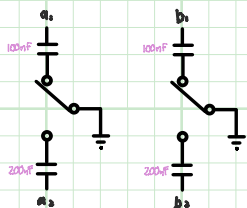
Circuit w/o Modulation



LEGEND:	
VIA: NE555	UIB: NE555
VCC:	VCC:
GND: 1	GND: 1



Circuit w/ Modulation ♡



LEGEND:

VIA: NESSS

VCC:

GND: 1

U2: 74HC08

VCC: 14, 13, 12

GND: 1, 2, 3

U1B: NESSS

VCC:

GND: 1